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Roll No

CE-221-CBCS

B.E., III Semester

Examination, June 2020

Choice Based Credit System (CBCS)

Fluid Mechanics

Time : Three Hours

Maximum Marks : 60

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Explain the terms Specific volume, specific gravity, surface tension and viscosity.
b) Explain with neat sketches the conditions of equilibrium for floating and submerged bodies.
2. Deduce an expression for Bernoulli's equation and also mention its applications.
3. a) Explain the concept of water hammer transmission of power.
b) Explain the boundary layer theory.
4. a) Explain the terms Stream lines, streak lines and path lines.
b) Explain various types of fluid flow.
5. a) Explain the surges in open channels.
b) Explain the concept of channel flow routing.

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6.
 - a) Differentiate between Gradually varied flow and Rapidly varied flow.
 - b) A trapezoidal channel having the side slope of 2H:3V carries water at $0.5\text{m}^3/\text{sec}$. If the bed slope of the channel is 1:1000, determine the optimum dimensions of the channel. Take Chezy's constant 'C' = 50.
7.
 - a) Explain the different types of drags in detail.
 - b) Explain the construction and working of a pelton wheel with a neat sketch.
8. Write short notes on any four of the following:
 - a) Specific energy
 - b) Bulk modulus of elasticity
 - c) Vorticity and circulation
 - d) Losses due to sudden expansion and contraction
 - e) Draft tube theory
